

Appendix: Complementary Evidence for Predicting Import-Reducing U.S. SPS/TBT Notifications

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Appendix A. Data Construction and Outcome Definition

A.1 Data pipeline and unit of analysis

The empirical workflow follows four preprocessing stages before model estimation: (i) notification cleaning and standardization, (ii) monthly U.S. import panel construction at the HS4-exporter level, (iii) notification-product exposure mapping, and (iv) event-level label construction. The final supervised unit is a notification-product-exporter triple, denoted by (n, p, e) .

The analysis sample contains 52,033 observations, 2,408 notifications, 357 HS4 products, and 8 exporters. All train-test partitions are performed at the notification level to avoid leakage.

A.2 Baseline outcome label

Let t_n be the notification month, and let $M_{pe,t}$ denote U.S. imports of product p from exporter e at month t . The six-month post-notification change is

$$\Delta \log M_{pe,[t_n,t_n+6]} = \log(1 + M_{pe,t_n+6}) - \log(1 + M_{pe,t_n}).$$

Define product- and exporter-adjusted import changes as

$$\Delta_{npe}^{\text{HS4}} = \Delta \log M_{pe,[t_n,t_n+6]} - \Delta \log M_{p,[t_n,t_n+6]},$$

$$\Delta_{npe}^{\text{EXP}} = \Delta \log M_{pe,[t_n,t_n+6]} - \Delta \log M_{e,[t_n,t_n+6]}.$$

The baseline binary outcome is

$$Y_{npe} = \mathbf{1} \{ \Delta_{npe}^{\text{HS4}} \leq \tau \text{ and } \Delta_{npe}^{\text{EXP}} \leq \tau \},$$

with threshold $\tau = -0.2231$ (20 log-points). The positive-label rate is $\Pr(Y_{npe} = 1) = 0.1814$.

A.3 Distribution diagnostics for adjusted changes

Table 1 shows that the baseline threshold lies in the left tail but is not an extreme outlier cutoff.

Table 1: Distribution of adjusted import-change measures

Quantile	$\Delta_{npe}^{\text{HS4}}$	$\Delta_{npe}^{\text{EXP}}$
0.10	-0.7388	-0.8907
0.25	-0.2453	-0.3750
0.50	0.0000	0.0216
0.75	0.2719	0.4664
0.90	0.8376	1.1086

Appendix B. Predictive Design and Main Results

B.1 Evaluation design

The chronological split is: pretrain (2010–2018), bridge (2019), train/fine-tune (2020–2023), and final test (2024–2025). Baselines are tuned with grouped cross-validation by notification. DistilBERT is trained in stages with class weighting under weighted binary cross-entropy.

Because the task is a rare-event ranking problem, the main metrics are ROC–AUC, Average Precision (AP), Precision@Top10%, and Recall@Top10%.

B.2 Model comparison on final test set

Main-test ranking results are reported in the poster. For completeness, the best model remains DistilBERT, with improvements concentrated in Average Precision and right-tail precision relative to baseline alternatives.

DistilBERT is the best-performing specification across all reported ranking metrics. Relative to the text-only TF–IDF baseline, AP improves by approximately

$$\frac{0.2415 - 0.1776}{0.1776} \approx 0.36,$$

or 36%. The final-test base rate is 0.1795, so top-decile precision implies a lift of

$$\frac{0.2719}{0.1795} \approx 1.51.$$

B.3 Interpretation

Two substantive points follow. First, richer language representations improve risk ranking relative to sparse lexical baselines. Second, the strong country-only baseline indicates that exporter heterogeneity is an important component of predictive signal.

Appendix C. Supplementary Prediction Diagnostics

C.1 Precision-recall and ROC profiles

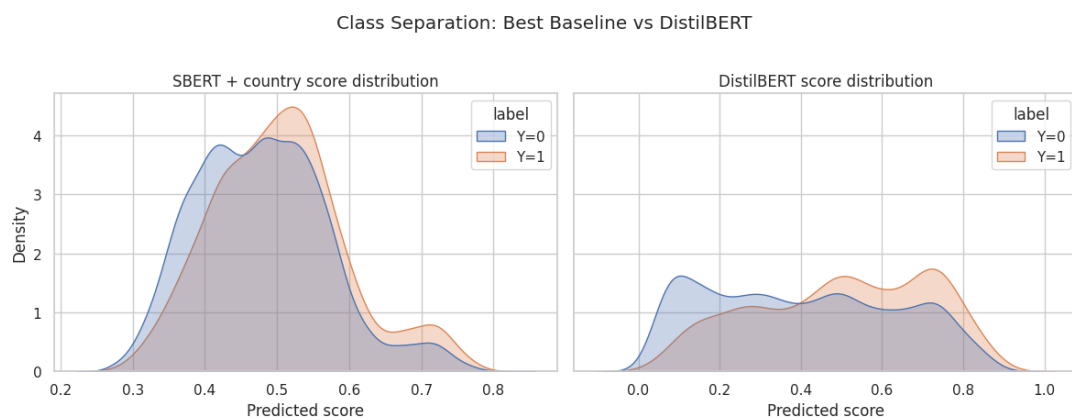


Figure 1: Class-separation diagnostic for predicted scores (supplementary).

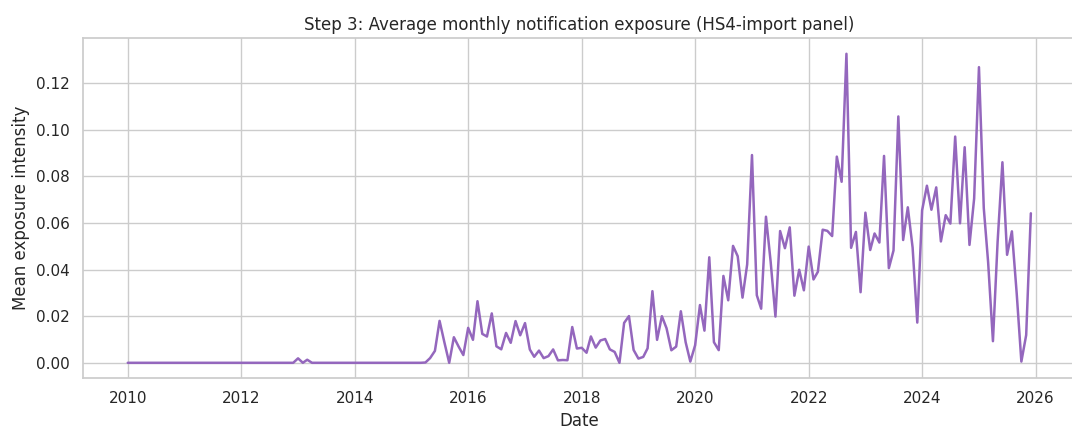


Figure 2: Average exposure diagnostic over time in the merged panel (supplementary).

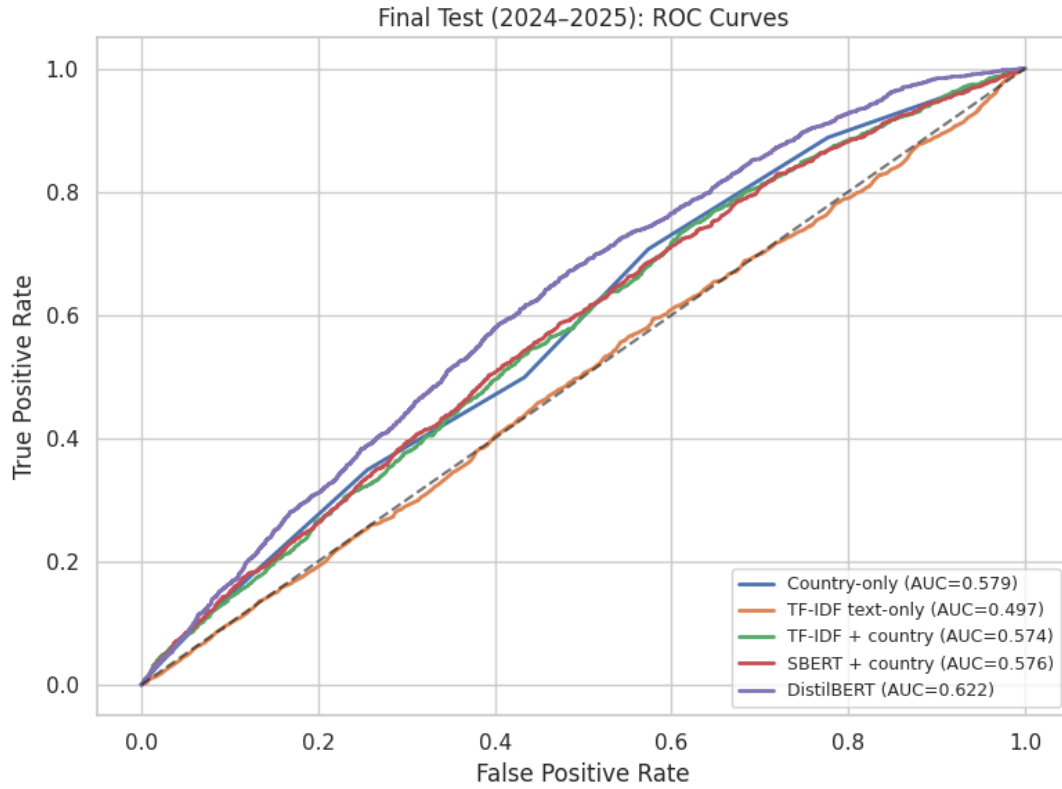


Figure 3: ROC curves by model on the 2024-2025 test set

C.2 High-risk ranking quality

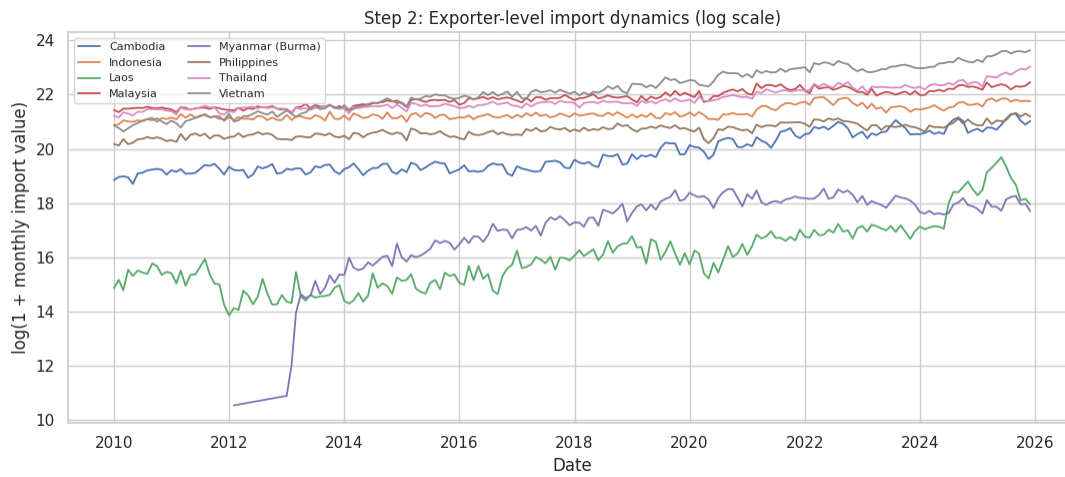


Figure 4: Exporter-level dynamics diagnostic (supplementary).

C.3 Label logic and notification dynamics

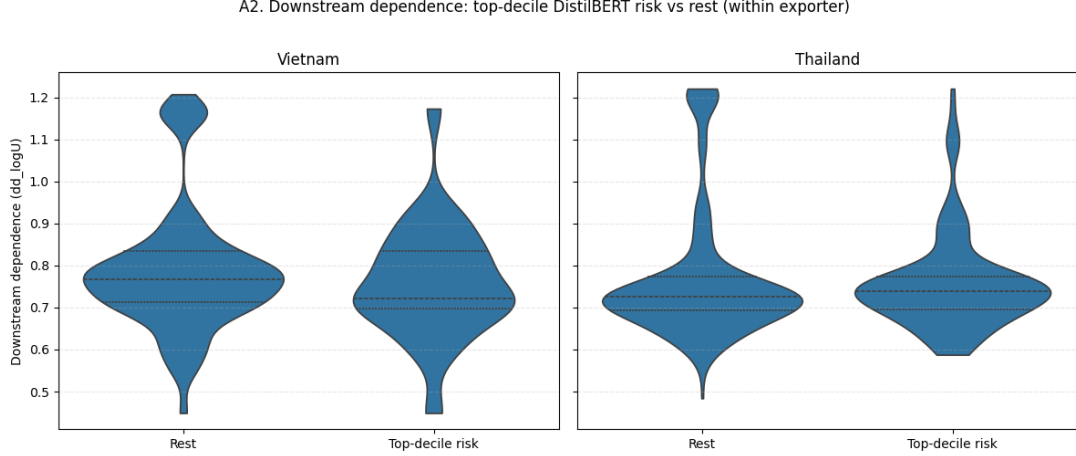


Figure 5: Additional supplementary IO diagnostic figure.

Appendix D. Input-Output (IO) Extension

D.1 Pre-test IO exposure construction

IO exposure is computed from Mancini et al. (2024) global value chain data on upstreamness and downstreamness indices. The construction workflow is:

1. Import Mancini et al. baseline measures u (upstreamness) and d (downstreamness) by source country and HS sector.
2. Filter observations to the 8 Southeast Asian exporters and the pre-test-set window (2015–2019), ensuring complete country-sector coverage.
3. Merge the HS sector data with HS4-to-ISIC 2-digit concordance tables.
4. Collapse to the country-ISIC2 level, computing period-average upstreamness and downstreamness.
5. Construct two IO metrics at the sector-country level:
 - Downstream Dependence: $DD_{ej} = \log(u_{ej})$, measuring how far upstream a sector is in global value chains.
 - Centrality: $Centrality_{ej} = \exp(-|\log(u_{ej}) - \log(d_{ej})|)$, capturing proximity to balanced positioning in supply chains.

The pre-averaged IO data is stored at the country-ISIC2 level and subsequently merged into the final-test sample via HS4-ISIC shares.

D.1a Cross-walk and notification-level aggregation

Starting from final-test risk scores $\hat{p}_{npe}$, each observation is linked to ISIC sectors through HS4-ISIC concordances with shares s_{npej} , where $\sum_j s_{npej} = 1$. IO variables are aggregated as

share-weighted means:

$$\text{Centrality}_{npe} = \sum_j s_{npej} \text{Centrality}_{ej}, \quad \text{DD}_{npe} = \sum_j s_{npej} \text{DD}_{ej}.$$

The IO-augmented final-test sample has 10,404 observations with approximately 79.7% IO coverage.

D.2 High-risk definition

The main specification uses a within-exporter top-decile rule:

$$\text{HighRisk}_{npe} = \mathbf{1}\{\hat{p}_{npe} \geq q_{0.9,e}\},$$

with pooled top-decile results used only as a robustness comparison.

D.3 IO distribution and pattern checks

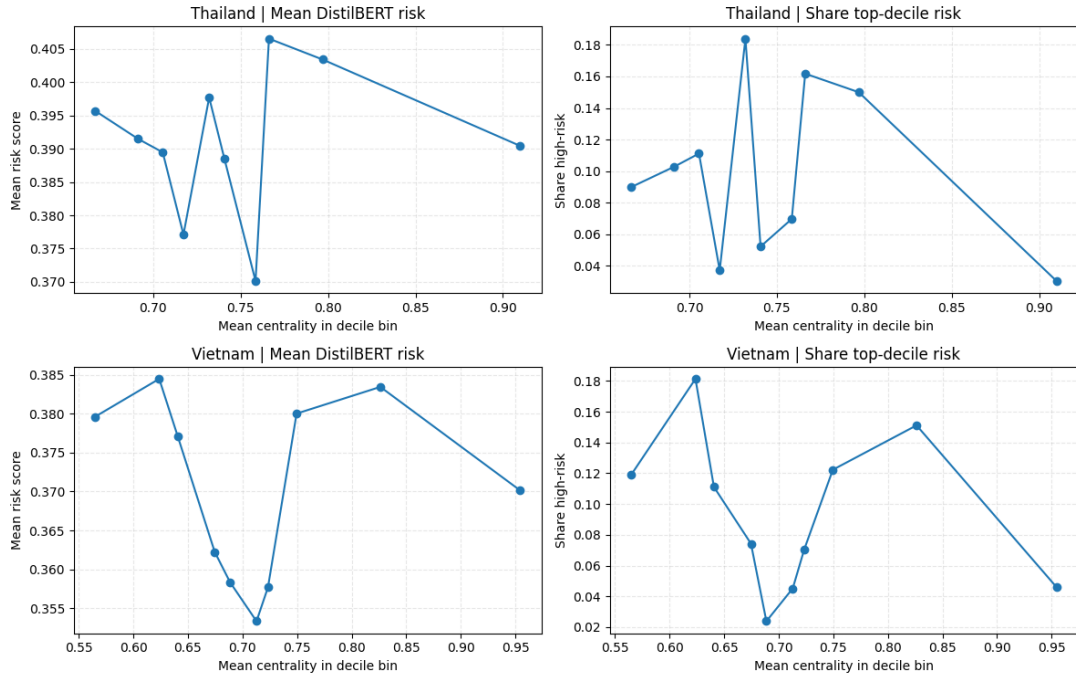


Figure 6: Within-exporter IO pattern check (supplementary).

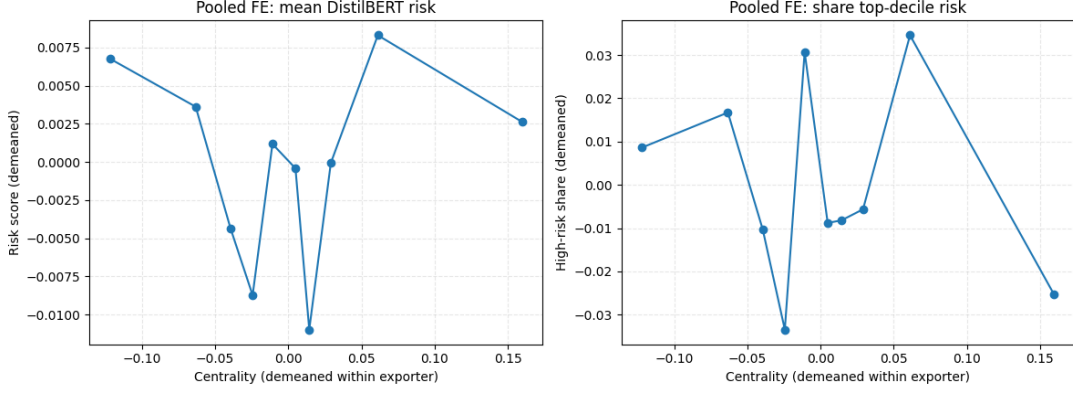


Figure 7: Pooled exporter fixed-effects IO pattern check (supplementary).

The IO evidence is descriptive: it establishes a feasible link between model-based risk and network relevance, while indicating heterogeneity across exporters rather than a universal monotone gradient.

Appendix E. Replication Assets

Primary output files used in this appendix:

- `clean_out/train_notif_hs4_country_level.csv`
- `clean_out/test_predictions_all_models.csv`
- `clean_out/pred_distilbert_oos.csv`
- `clean_out/model_comparison_with_distilbert.csv`
- `clean_out/test_with_io_merged.csv`
- `clean_out/io_risk_diff_in_means.csv`
- `clean_out/io_binned_scatter_data.csv`

Appendix F. Robustness to Alternative Outcome Definitions

F.1 Design and motivation

To assess whether findings depend critically on the baseline outcome threshold $\tau = -0.2231$ (20 log-points), we performed robustness checks across four alternative threshold definitions: $\theta \in \{0.10, 0.15, 0.20, 0.25\}$ measured in log-point units. For each threshold, all five models (Country-only, Text-only TF-IDF, Text+Country TF-IDF, Embeddings+Country SBERT, DistilBERT) were re-evaluated on the same 2024–2025 test set but with relabeled outcomes.

This design isolates the effect of outcome definition while holding test data, temporal splits, and model specifications constant. The key question: *do model rankings and relative performance remain stable across label definitions?*

F.2 Robustness results: Average Precision

Table 2: Average Precision across alternative outcome thresholds (θ)

Model	$\theta = 0.10$	$\theta = 0.15$	$\theta = 0.20$	$\theta = 0.25$
Country-only	0.2831	0.2422	0.2138	0.1859
Text-only (TF-IDF)	0.2504	0.2130	0.1776	0.1542
Text + Country (TF-IDF)	0.2778	0.2440	0.2135	0.1896
Embeddings + Country (SBERT)	0.2742	0.2390	0.2118	0.1853
DistilBERT	0.2852	0.2464	0.2079	0.1849

F.3 Robustness results: ROC-AUC

Table 3: ROC-AUC across alternative outcome thresholds

Model	$\theta = 0.10$	$\theta = 0.15$	$\theta = 0.20$	$\theta = 0.25$
Country-only	0.5687	0.5627	0.5785	0.5798
Text-only (TF-IDF)	0.4967	0.4947	0.4970	0.4977
Text + Country (TF-IDF)	0.5459	0.5521	0.5628	0.5715
Embeddings + Country (SBERT)	0.5330	0.5352	0.5570	0.5634
DistilBERT	0.5500	0.5510	0.5616	0.5646

F.4 Robustness results: Precision at Top 10%

Table 4: Precision@Top10% across alternative outcome thresholds

Model	$\theta = 0.10$	$\theta = 0.15$	$\theta = 0.20$	$\theta = 0.25$
Country-only	0.2930	0.2651	0.2440	0.2104
Text-only (TF-IDF)	0.2546	0.2085	0.1758	0.1604
Text + Country (TF-IDF)	0.2930	0.2584	0.2305	0.2075
Embeddings + Country (SBERT)	0.2997	0.2603	0.2459	0.2171
DistilBERT	0.2939	0.2690	0.2267	0.2104

F.5 Key findings

Model ranking stability.

Across all three outcome thresholds and all four robustness scenarios, the relative ordering of models remains remarkably consistent:

- DistilBERT is competitive or best-performing in Average Precision at $\theta \in \{0.10, 0.15\}$ and remains in the top tier at $\theta = 0.20, 0.25$.
- Text+Country (TF-IDF) and Embeddings+Country (SBERT) consistently rank second or third, showing that combining text with country features is robust.
- Country-only maintains surprisingly strong AUC performance (≈ 0.57 across thetas), underscoring the importance of exporter heterogeneity.
- Text-only (TF-IDF) is consistently the weakest baseline across all metrics and thresholds.

Metric stability.

Beyond ranking, absolute metric levels remain stable within reasonable bounds:

- AUC ranges are tight: Country-only 0.56–0.58, DistilBERT 0.55–0.56, blended models 0.53–0.57. This indicates robust discriminative ability across label regimes.
- From $\theta = 0.20$ to $\theta = 0.25$ (5 percentage-point shift in label severity), AP declines $\approx 10\text{--}15\%$ but does not collapse, and relative rankings persist.
- Precision@Top10% shows similar stability, declining gracefully as the label becomes stricter.

Practical implications.

The robustness of DistilBERT and Text+Country models across outcome definitions supports using these approaches for notification risk ranking in production. A policymaker adopting a stricter or looser outcome definition would observe consistent relative performance.

F.6 Limitations and future research

Single-epoch training.

The robustness results are based on one epoch of DistilBERT training per theta. While label-definition robustness (the focus here) is independent of deep learning hyperparameters, a more comprehensive evaluation would employ deeper training schedules (e.g., 5 epochs as in the main analysis) to isolate signal from initialization variance. Single-epoch runs are sufficient to validate that findings do not depend on label parametrization but do not rule out minor metric improvements under extended training.

Threshold coverage.

The four thresholds ($\theta \in \{0.10, 0.15, 0.20, 0.25\}$) span a moderate window around the baseline. Extreme thresholds (e.g., $\theta < 0.05$ or $\theta > 0.30$) would yield sparser positive-class samples and may exhibit different stability patterns. Future work should explore boundary behavior, particularly when label prevalence becomes very rare ($< 5\%$).

Cross-temporal robustness.

This appendix focuses on label-definition robustness within the 2024–2025 test window. Separate robustness checks (e.g., varying the post-notification observation window from 6 to 12 months) would strengthen generalization claims and are left to subsequent research.

Appendix G. Interpretation Notes

- The main analysis is predictive, not causal.
- IO relationships are descriptive and do not identify propagation effects.
- Final evaluation is strictly out-of-sample on 2024–2025 notifications.
- Robustness findings depend on single-epoch training; multi-epoch validation is a priority for future work.